Important Notes

1. All geometry was adjusted to standard slicer settings and basic PLA, you can add walls and infill for stronger parts, using PETG will also enforce parts.

2. In most cases flat bolts can be swapped to countersunk bolts.

3. We used TPU by bambulab for tires, they should print in 7H, standard settings for TPU in bambulab slicer are recommended.

4. Main grey color used in videos is light grey 10104 PLA by bambu.

5. We highly recommend using 2F modules after base and before roof, when you are using modifications.

6. Holes for casters are M8, make sure to check it before purchasing them.

VAULT



3D Printed Modular Storage System

Instructions Manual

Hardware List

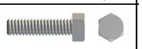
DRAWERS

1.	M4 14MM bolts	x4	*per floor*	
2.	M6 50MM bolts	x2	*per roof handle*	

SUSTAINERS

1.	M4 10MM bolts	x8	*per module*	
2.	M4 14MM bolts	x16	*per module*	
3.	M4 20MM bolts	x7	*per module*	

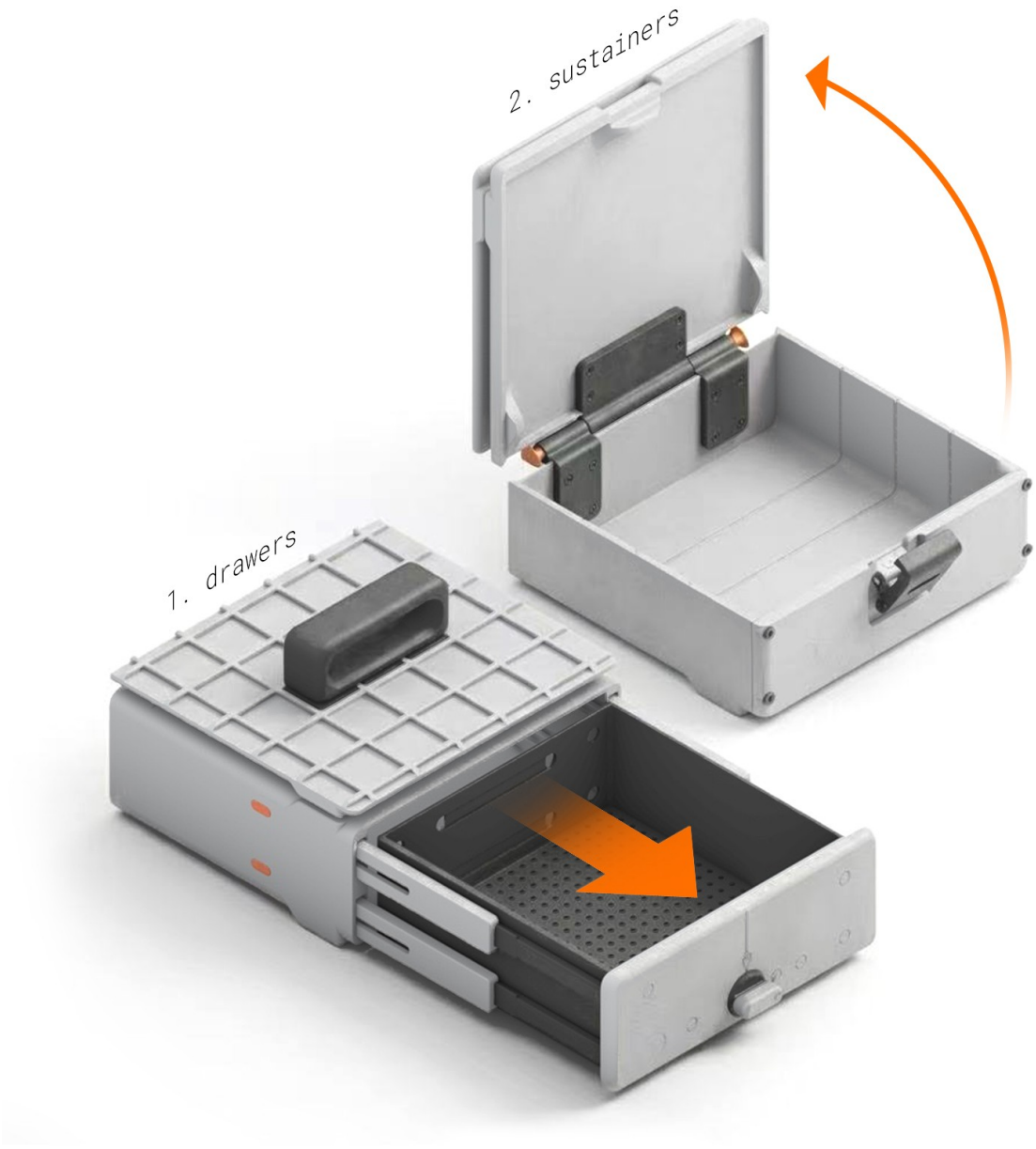
MODIFICATIONS

1.	M6 30MM bolts	x4	*per pull-out handle*	
2.	M6 35MM bolts	x2	*per pull-out handle*	
3.	M6x8x8 heat insert	x2	*per pull-out handle*	
4.	M4 60MM bolts	x8	*per static handle*	
5.	M4 40MM bolts	x4	*per pegboard board*	
6.	M4 60-100MM bolts	x4	*per backbone board*	
7.	M16 504MM rod	x1	*per XL base*	
8.	M16 414MM rod	x1	*per L base*	
9.	M16 338MM rod	x1	*per M base*	
10.	M16 248MM rod	x1	*per S base*	
11.	M16 N-MM rod	x2	*per pull-out handle*	
12.	M4 25MM bolts	x6	*per wheel*	
13.	M6 90MM bolts	x2	*per foot*	
14.	6205ZZ bearing	x1	*per wheel*	
15.	SCSAW stopper	x1	*per wheel*	

Five Ways Of Using - VAULT

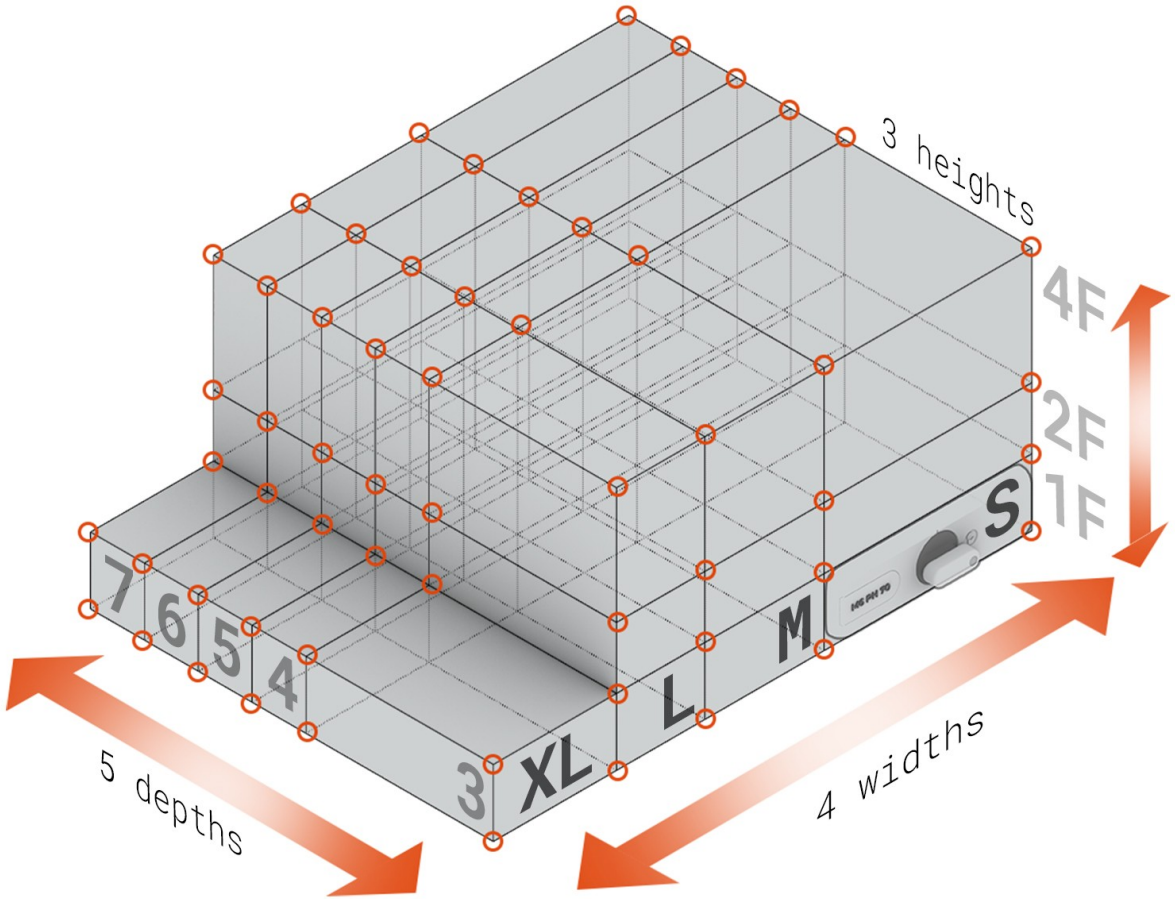


Two Types of - VAULT



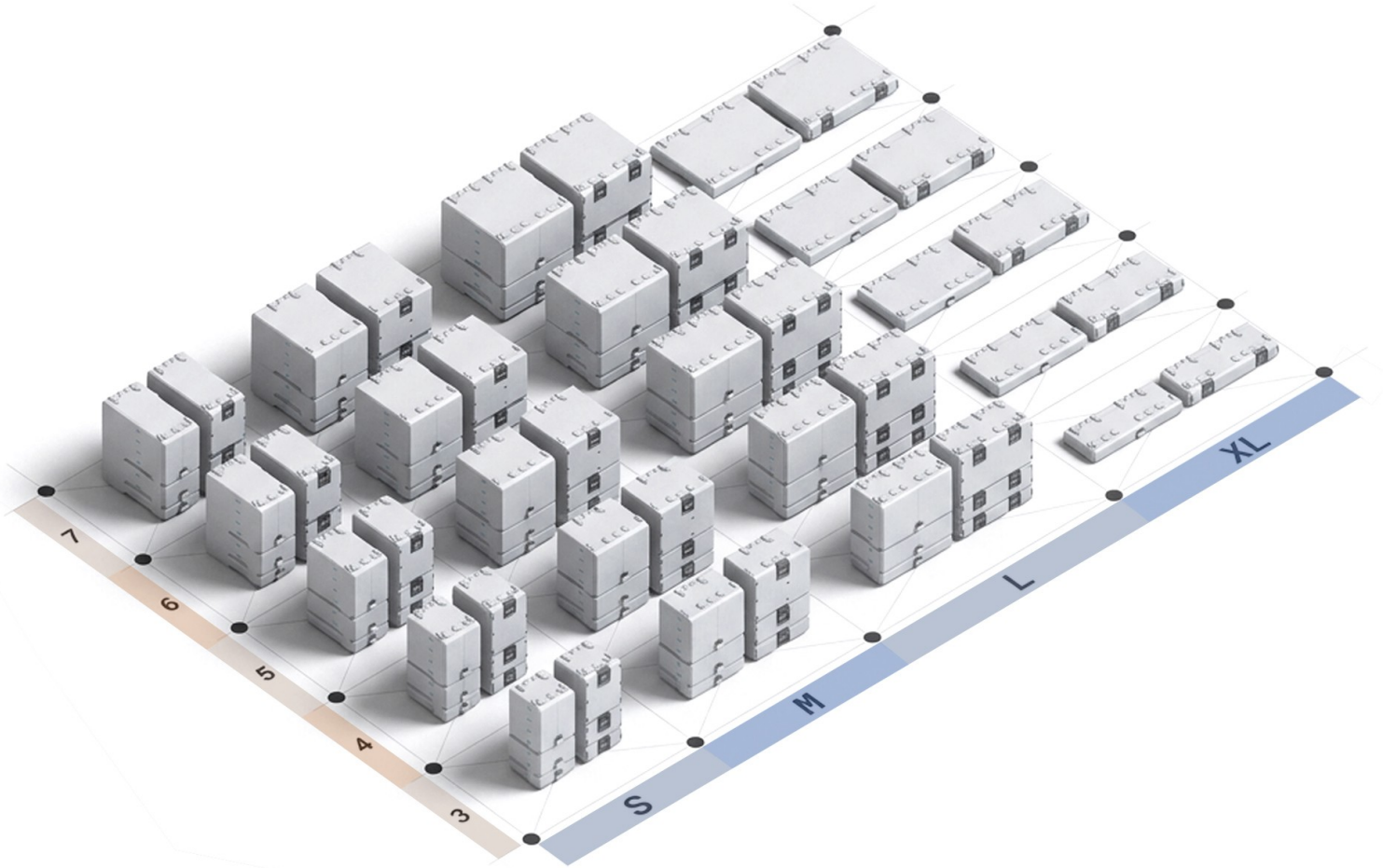
*archive is split into two parts as well

S, M, L, XL - 50 Size Variants

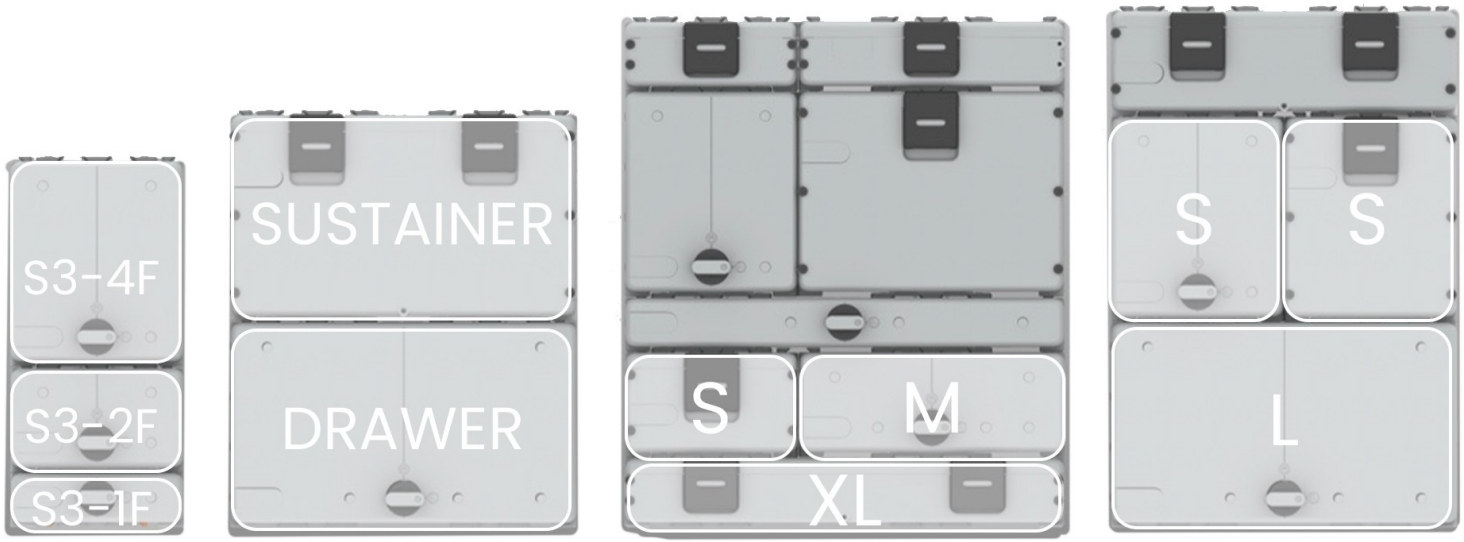


*depths are determined by gridfinity units (42mm x3, 42mm x4..)
*widths are determined by gridfinity units and printer sizes
*1F, 2F and 4F are the heights, no 3F to avoid gaps, everything divides by 2

160 Module Variants

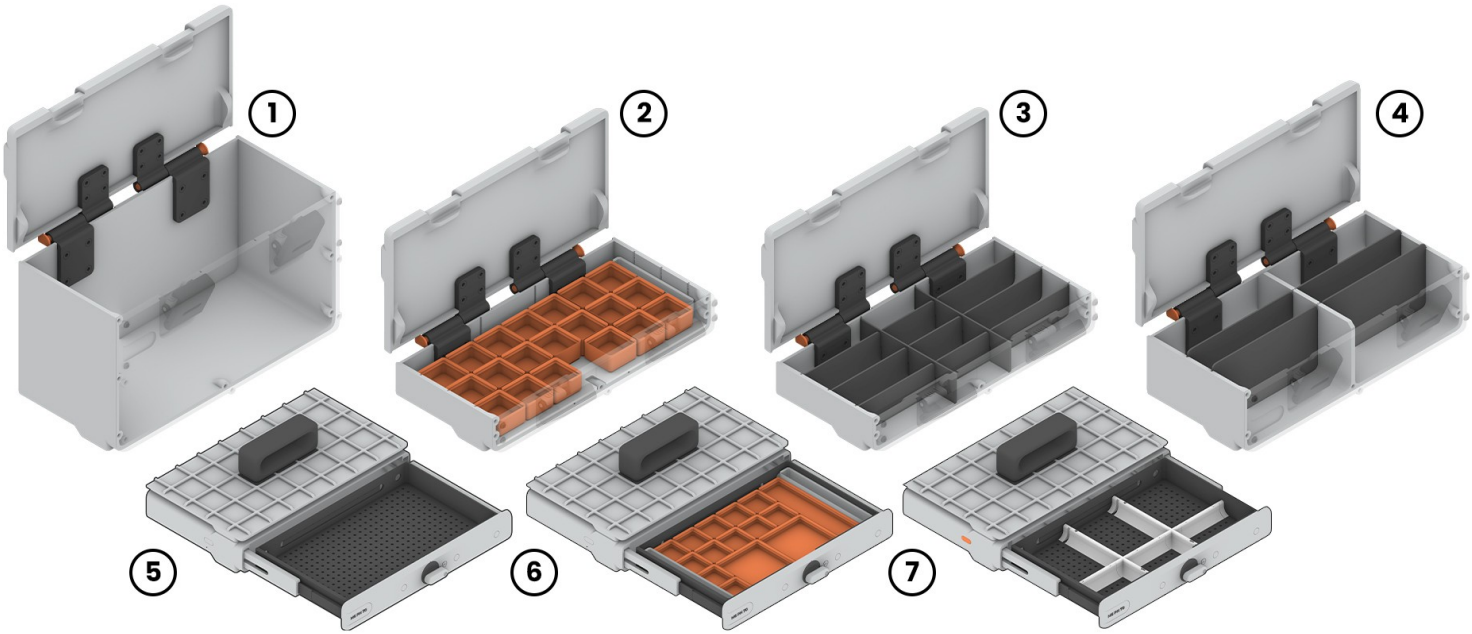


Stacking Options



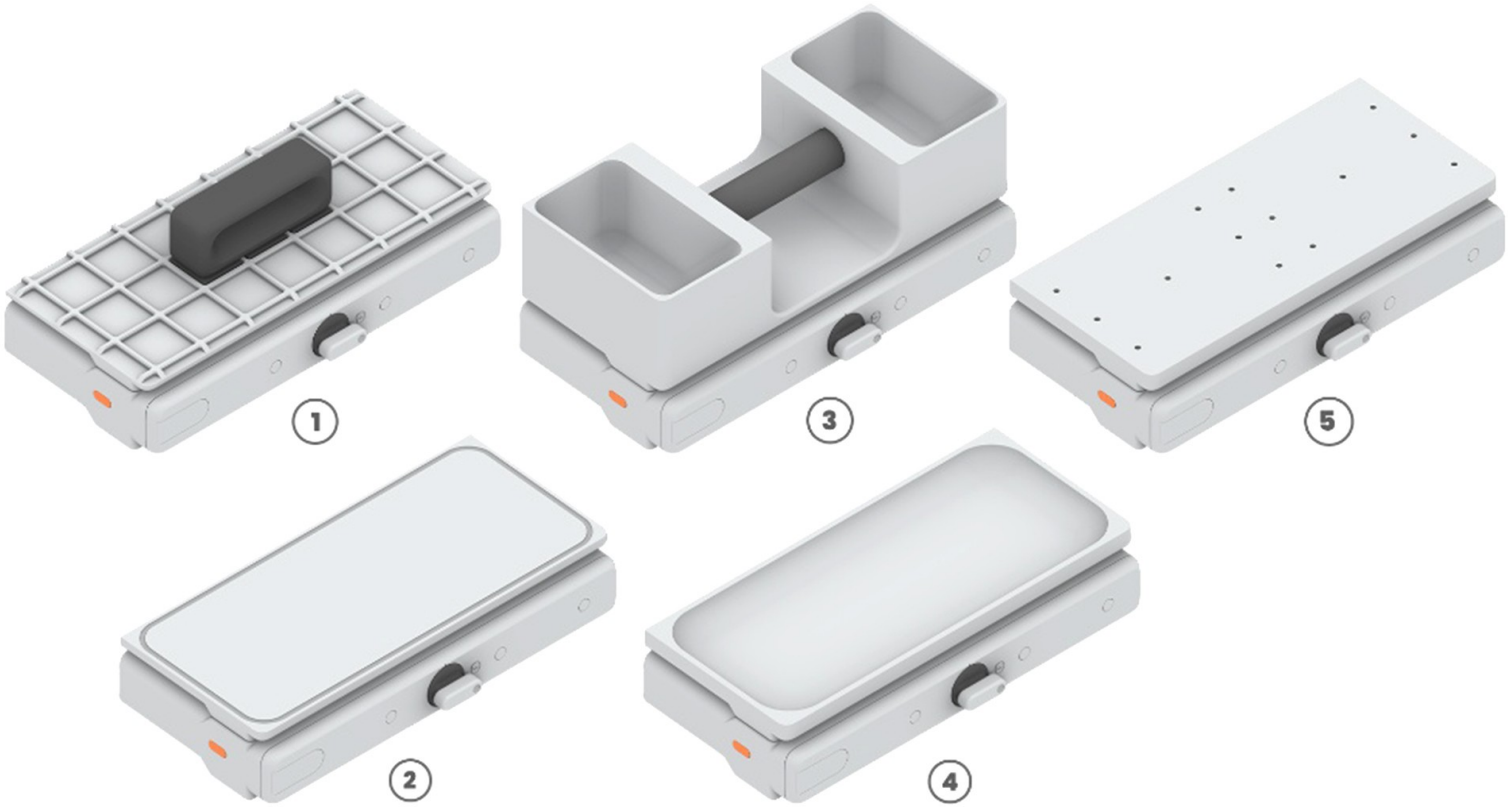
1. Modules with similar depth and width stack on each other.
2. SUstainers and Drawers can stack as well.
3. XL in width, equals to S+M, so they stack as well.
4. L in width, equals to S+S. they stack too.

Interiors



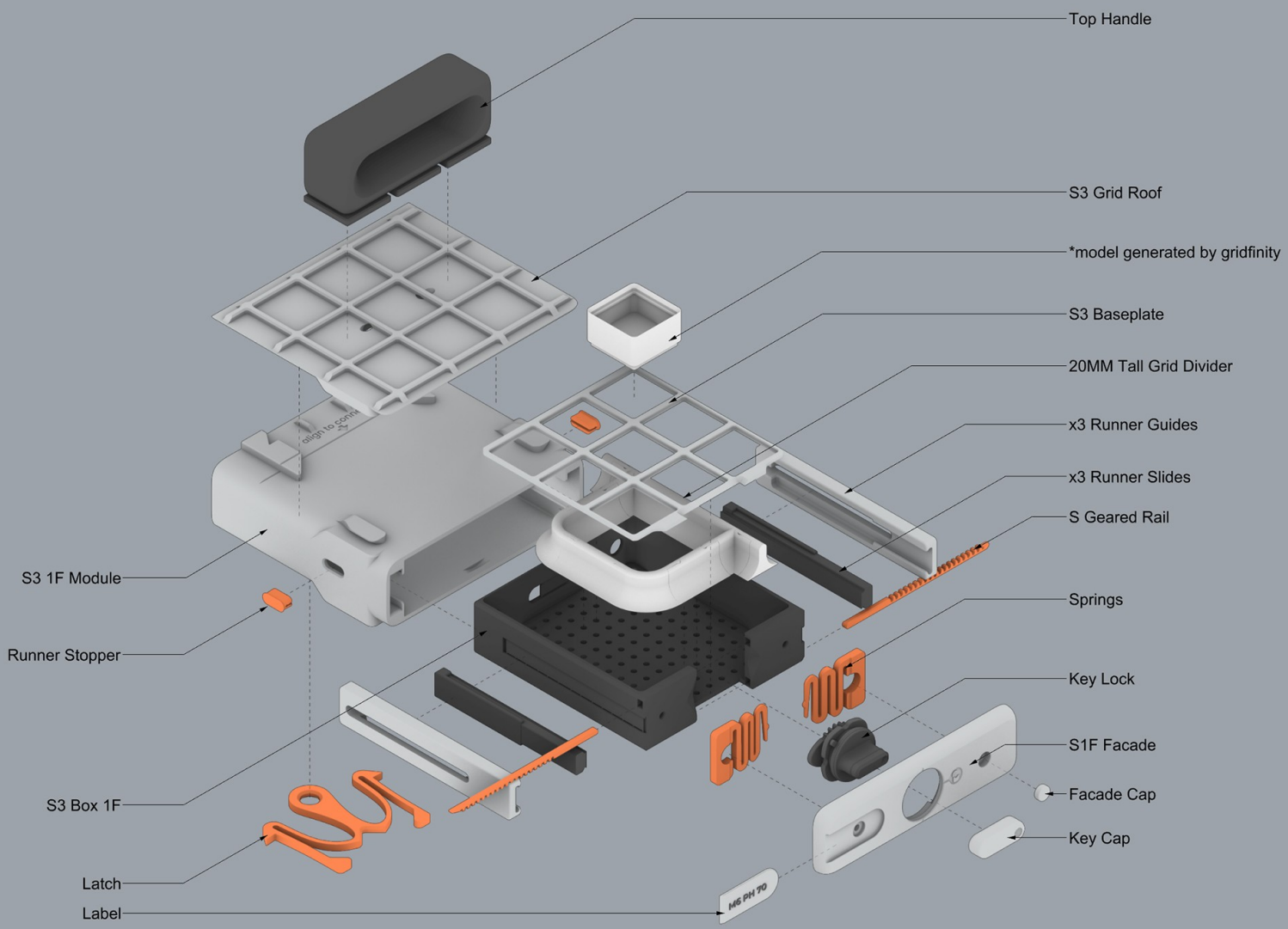
1. flat sustainer 2. gridfinity base 3. dividers sustainer 4. sealed sustainer
5. empty drawer 6. gridfinity base 7. grid dividers

Roof Types

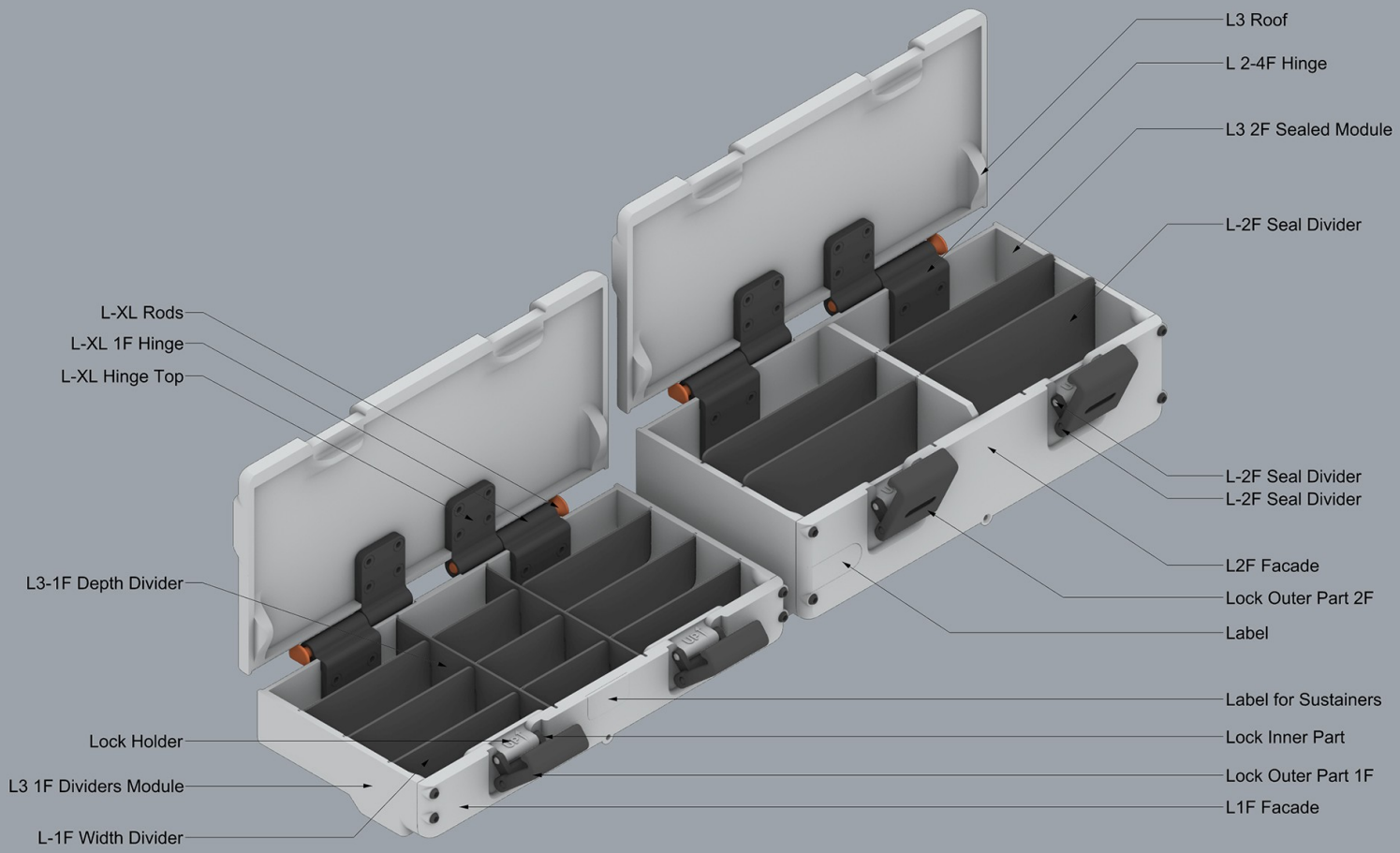


1. gridfinity 2. table 3. heavyweight 4. bowl 5. under table mount

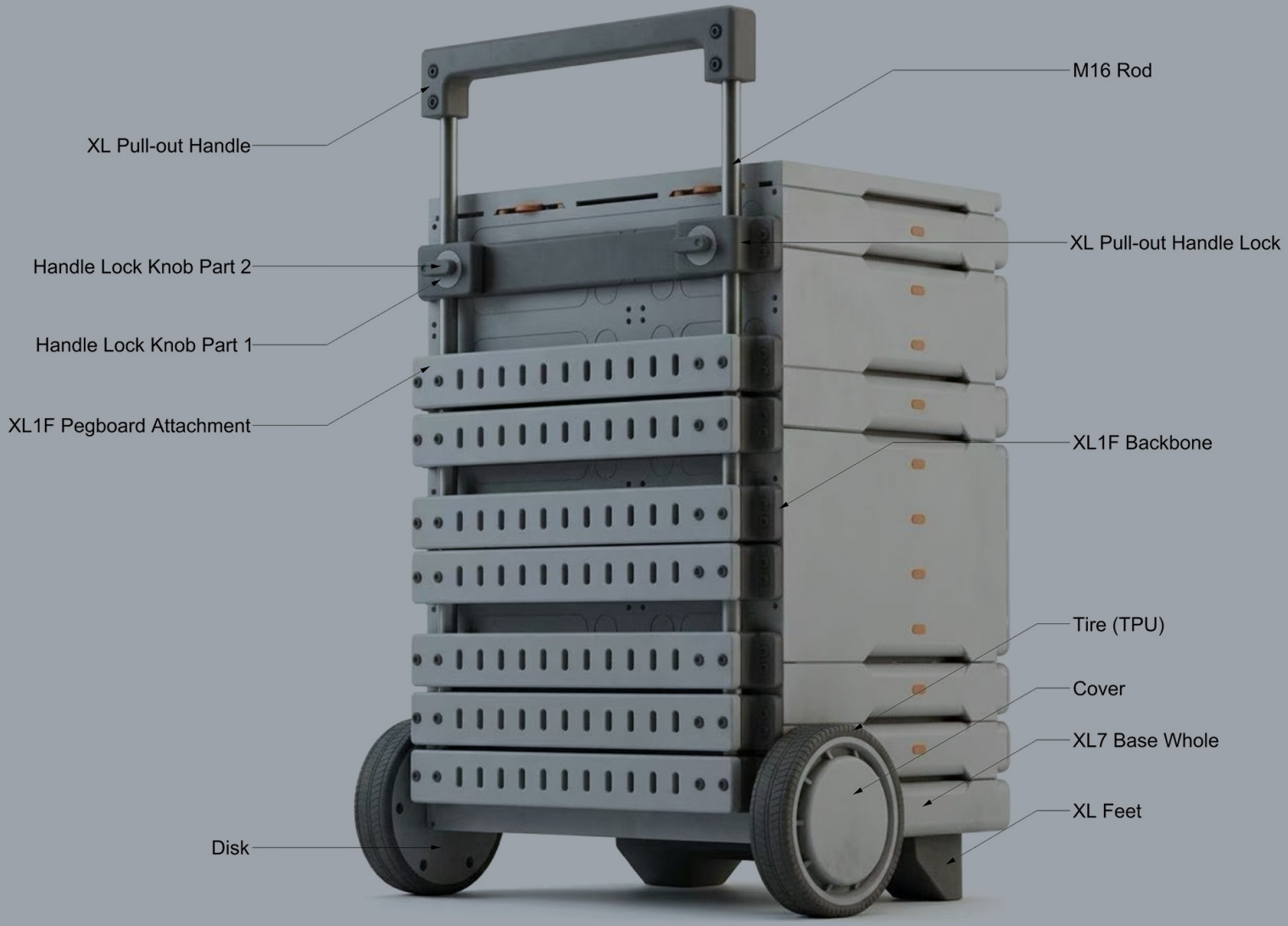
Drawers Naming



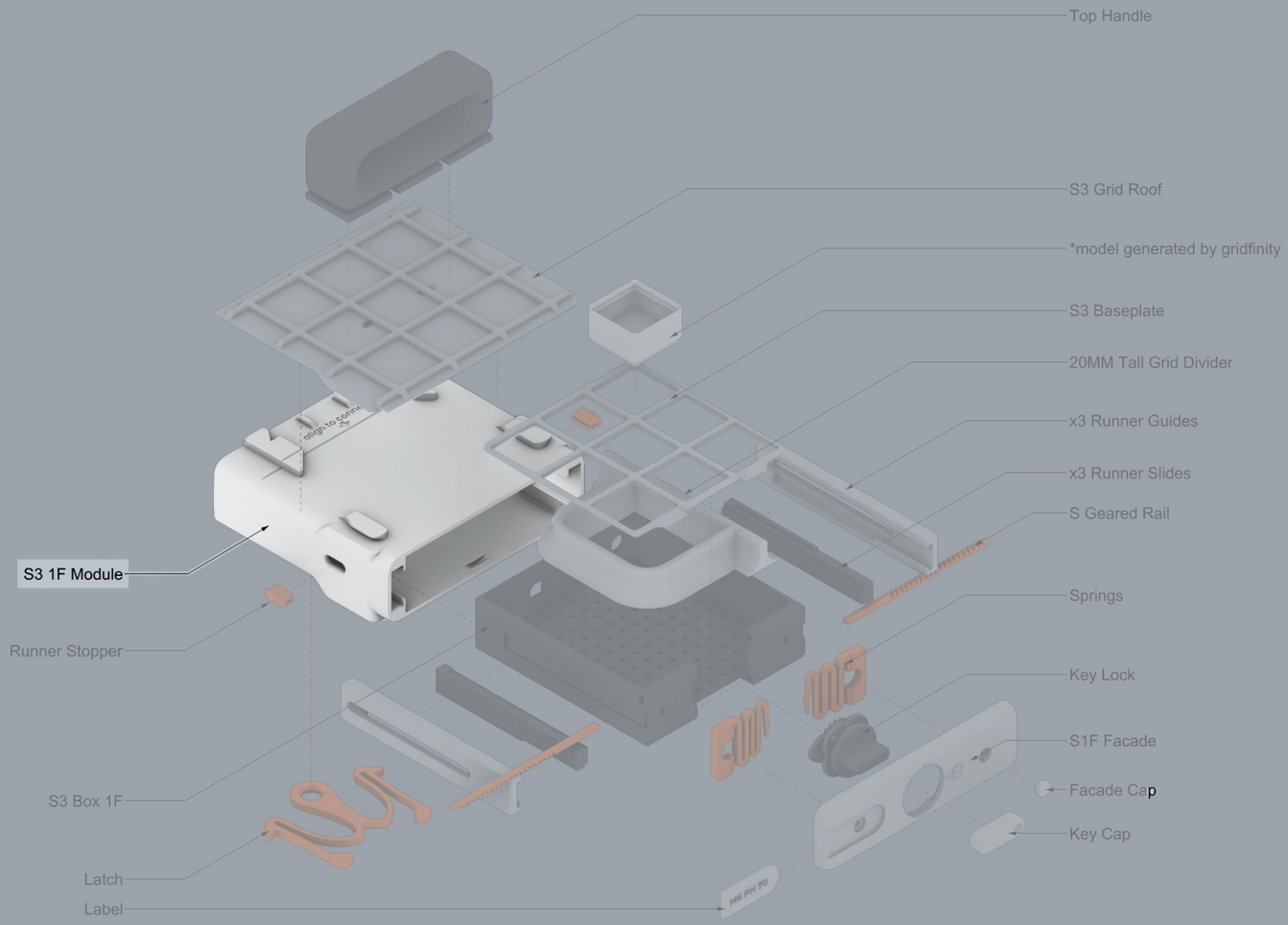
Sustainers Naming

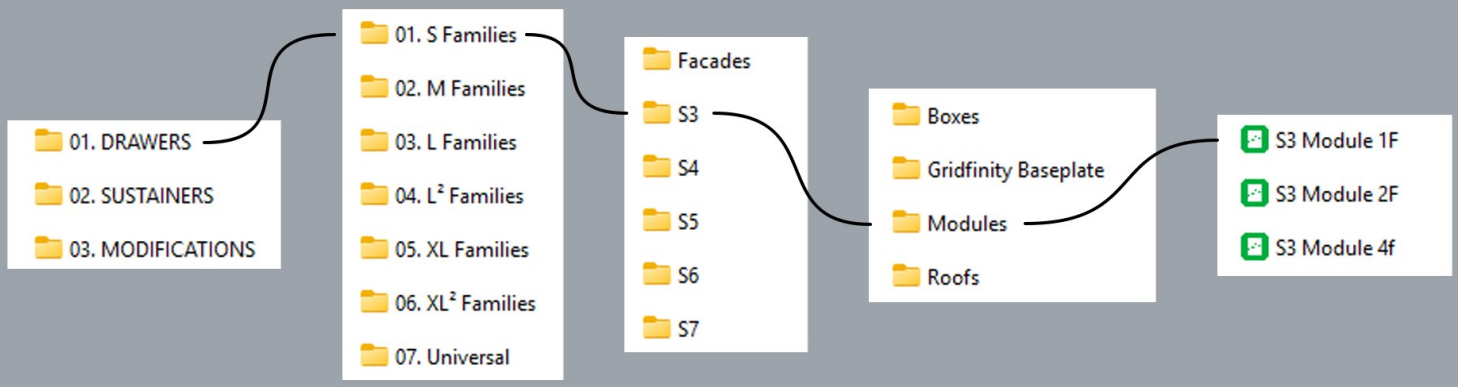


Modifications Naming

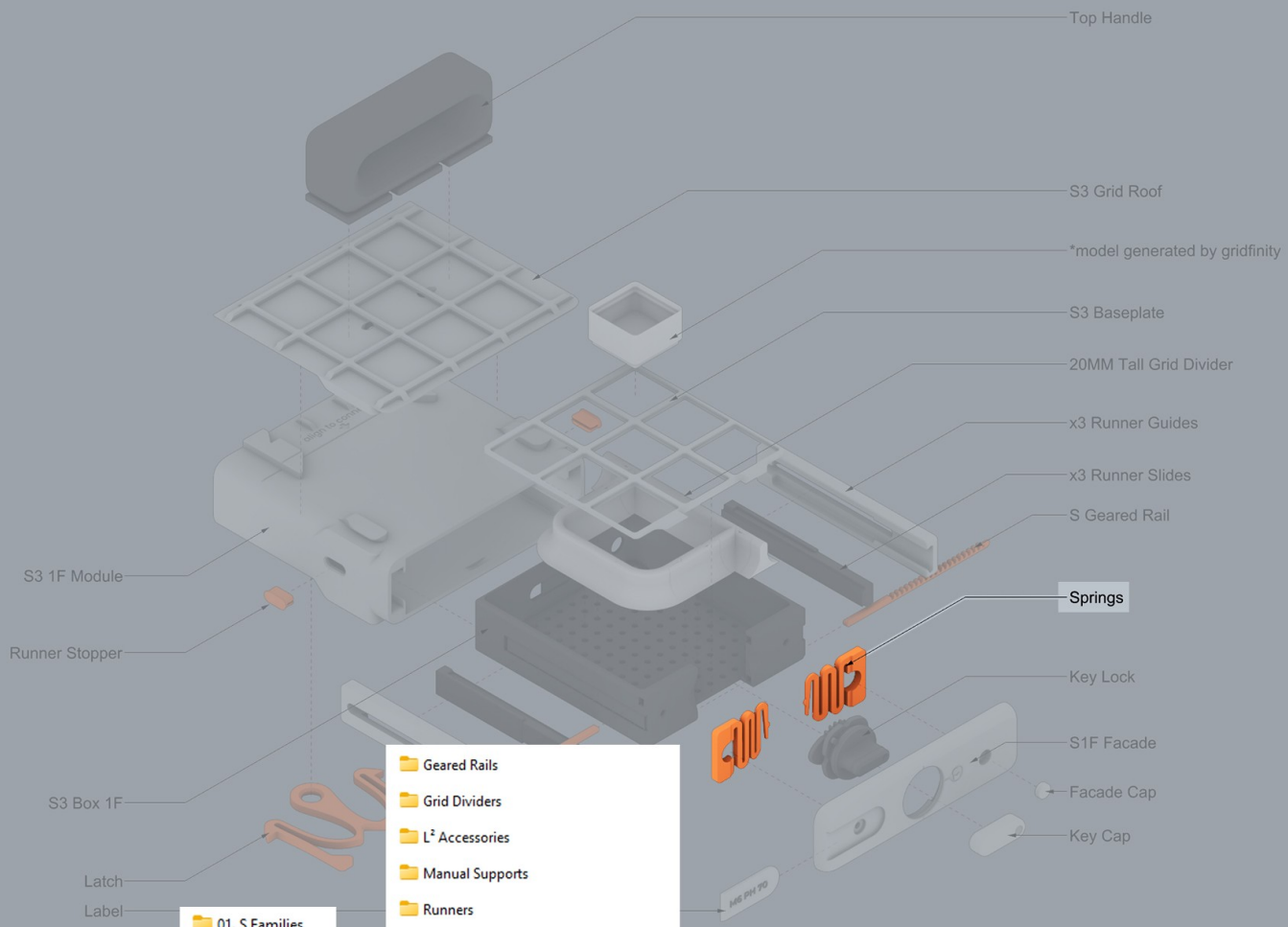


Working With Archive



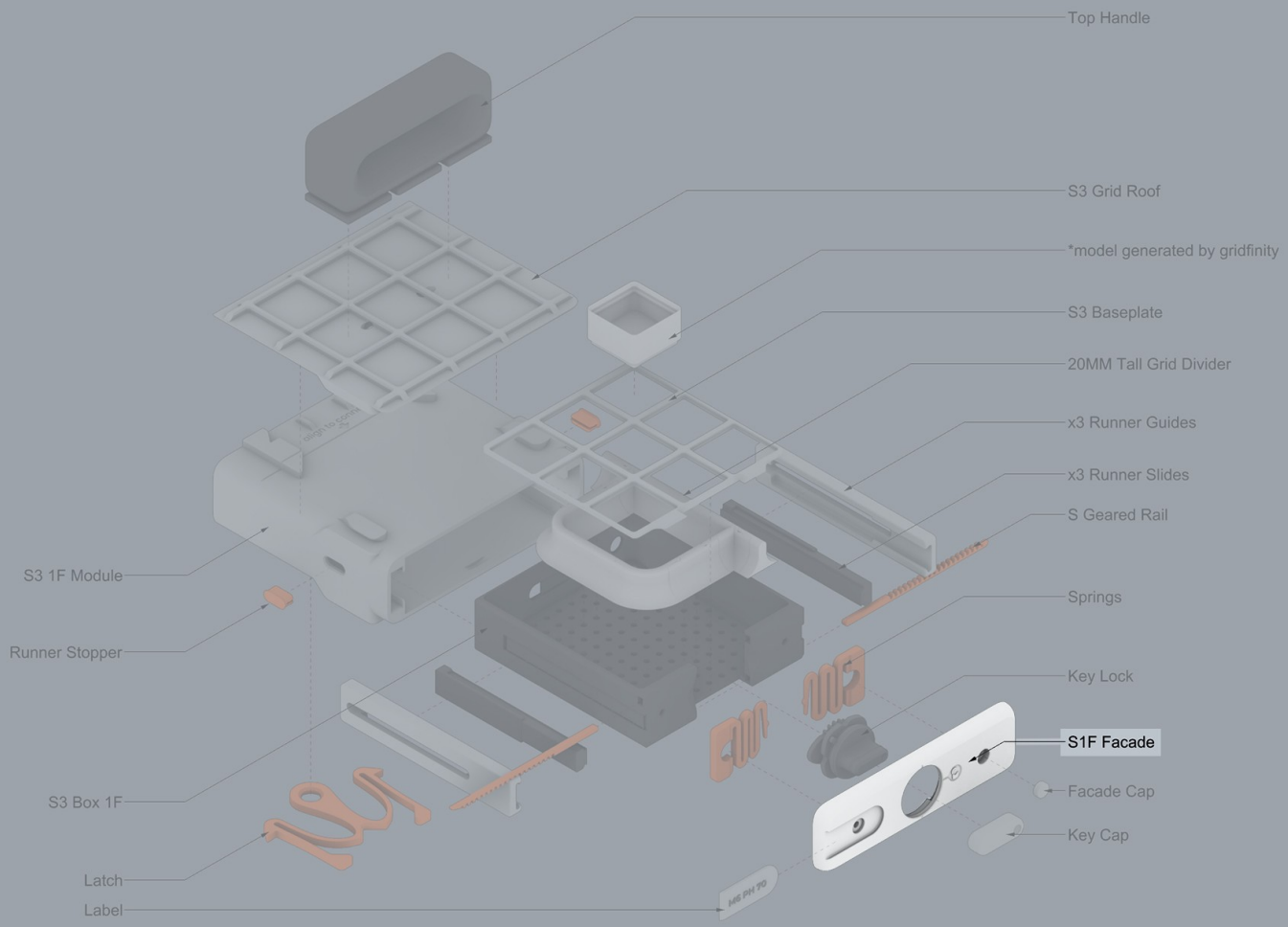


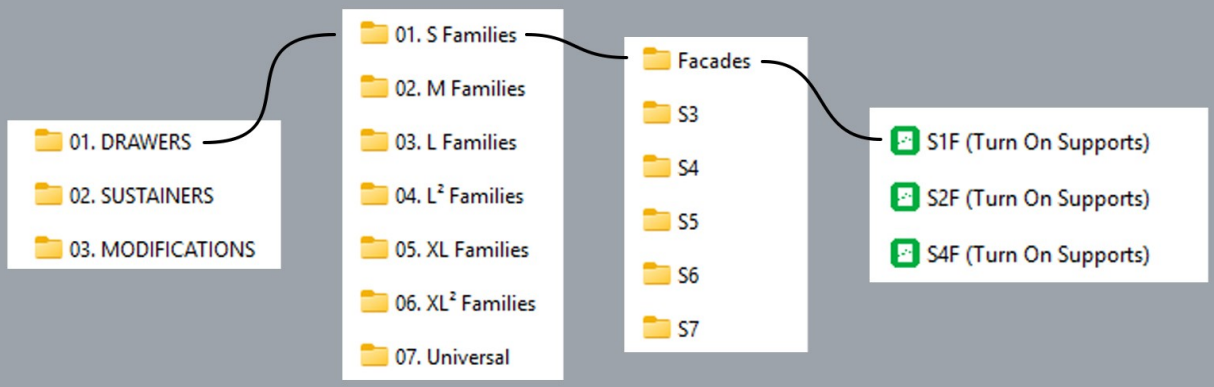
Working With Archive



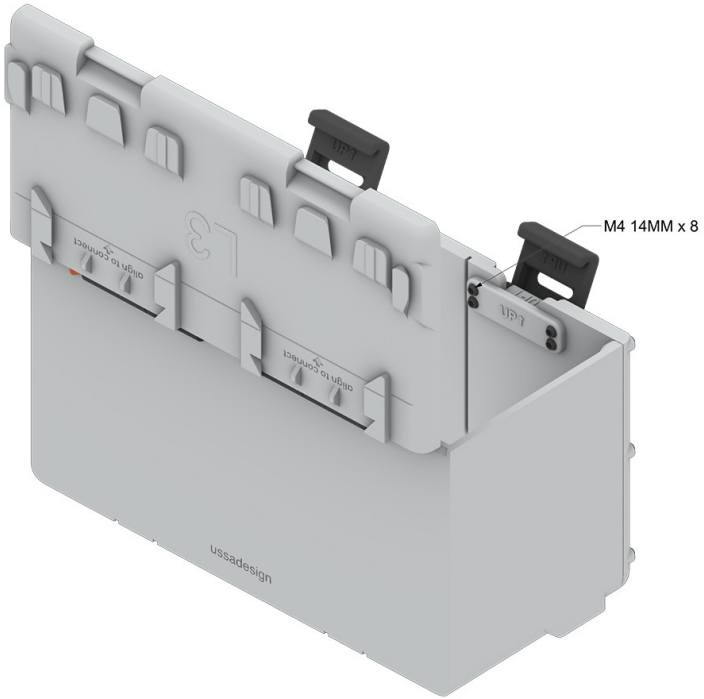
- 01. DRAWERS
- 02. SUSTAINERS
- 03. MODIFICATIONS
  - 01. S Families
  - 02. M Families
  - 03. L Families
  - 04. L² Families
  - 05. XL Families
  - 06. XL² Families
  - 07. Universal
    - Geared Rails
    - Grid Dividers
    - L² Accessories
    - Manual Supports
    - Runners
    - Facade Cap
    - Heawyweight Roof Handle
    - Heawyweight Roof Insert
    - Hidden Key (Turn On Supports)
    - Hidden Key Part 2 (Turn on Supports)
    - Key (Turn On Supports)
    - Key Cap
    - Label
    - Latch
    - Springs
    - Top Handle (Turn On Supports)

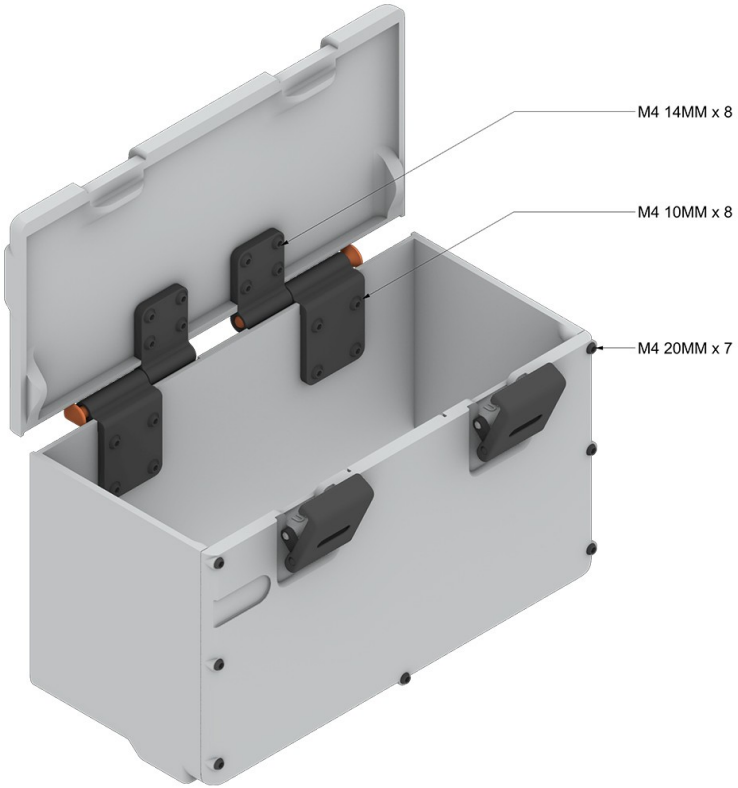
Working With Archive





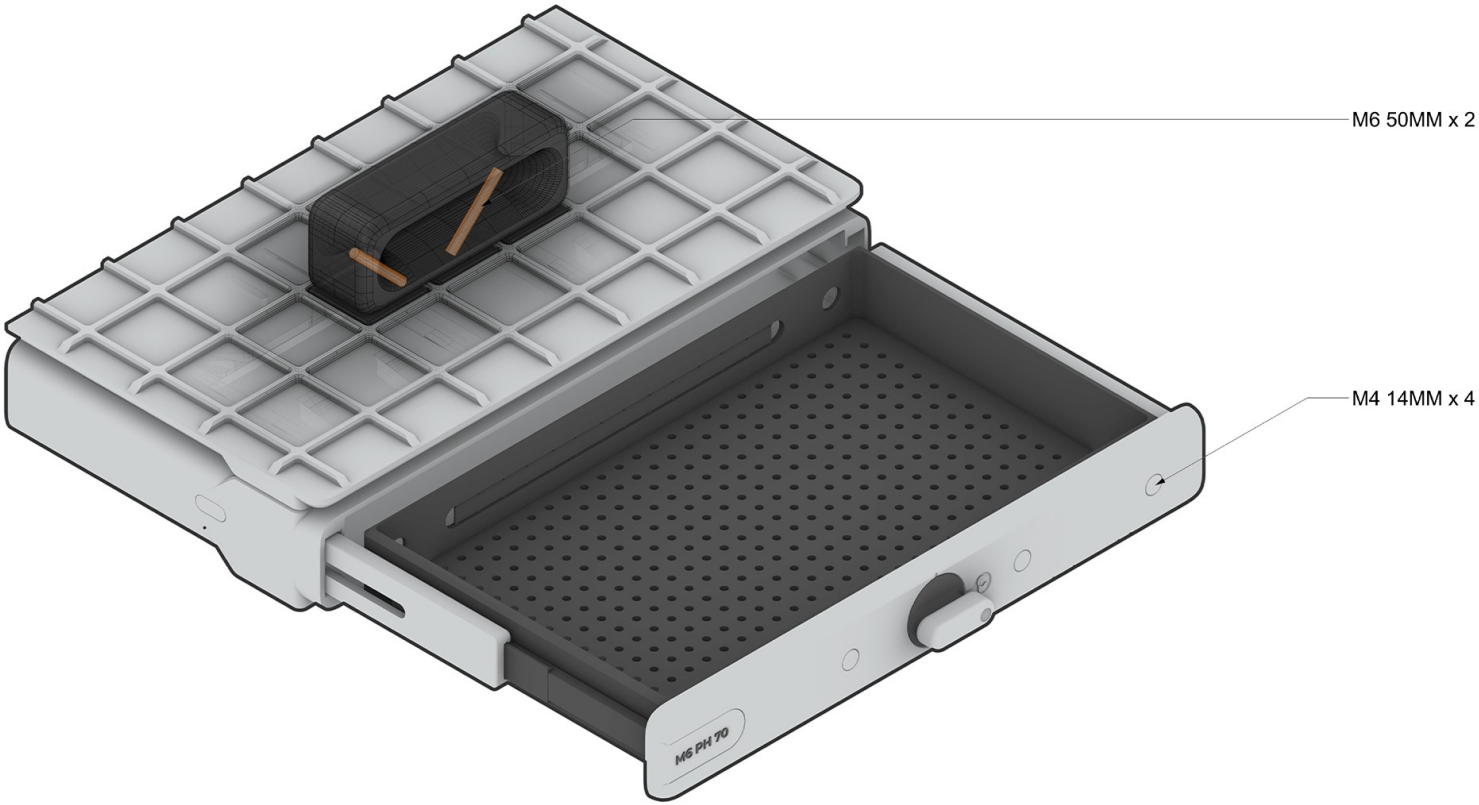
Required Hardware - Sustainers





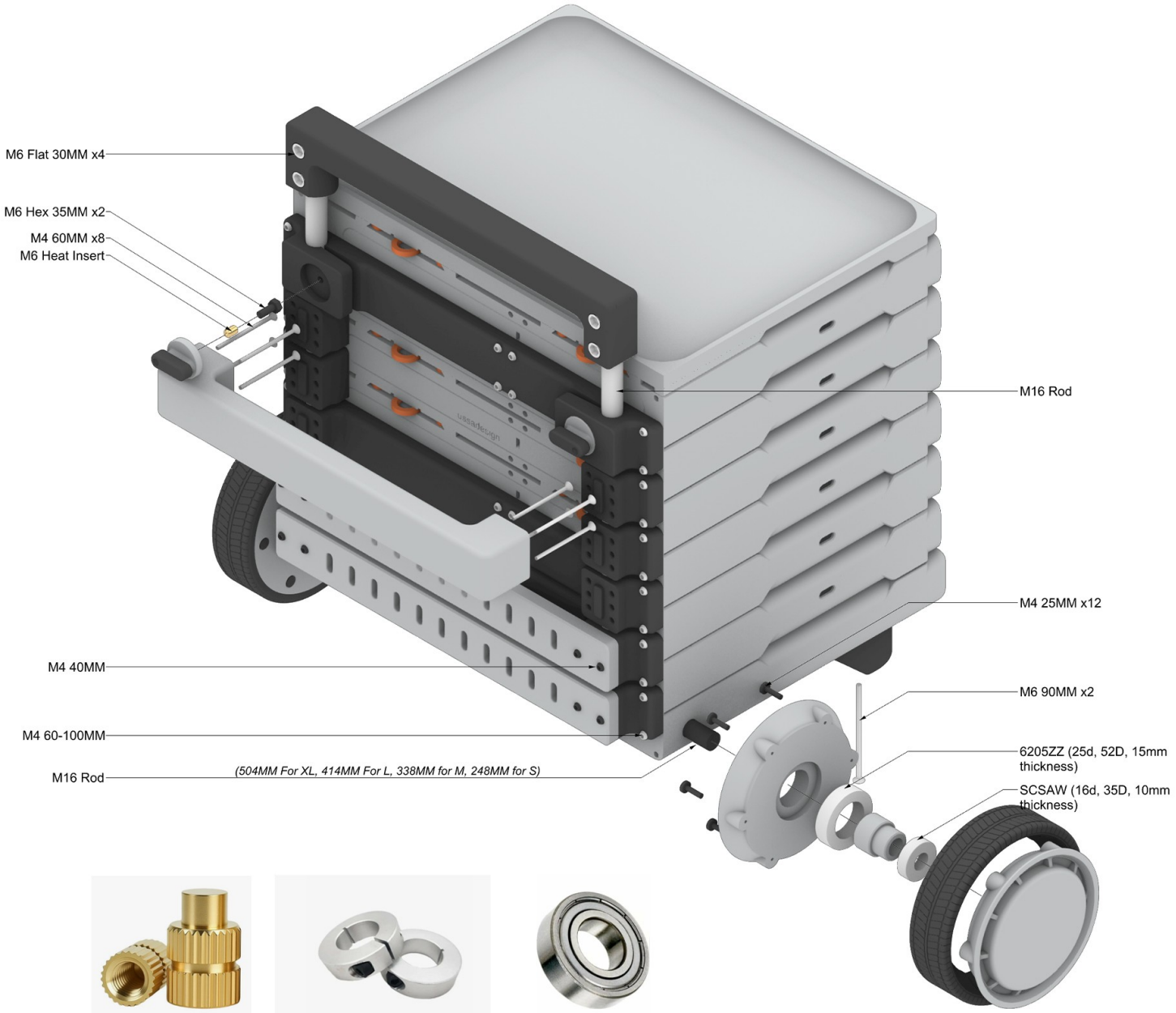
*x16 M4 14mm bolts, x8 M4 10mm bolts, x7 M4 20mm bolts - per module

Required Hardware - Drawers



*x4 M4 14mm bolts - for 1F, x6 for 2F/4F

Required Hardware - Backbone

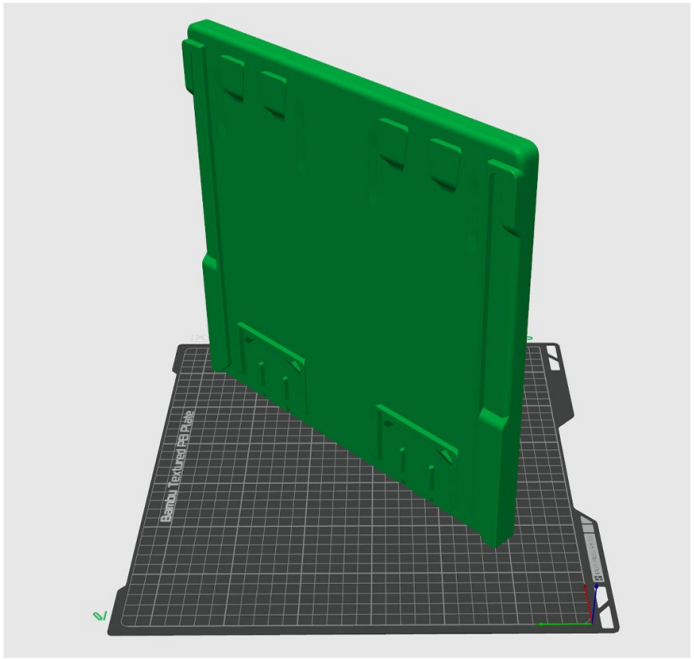


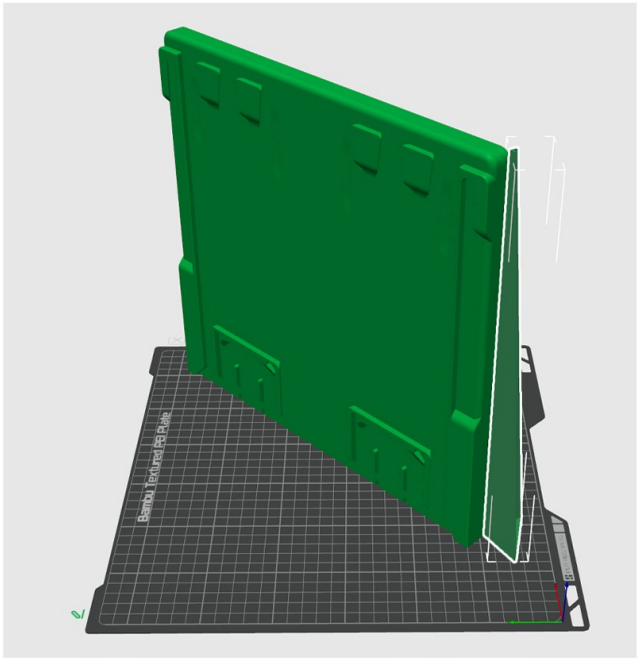
heat insert

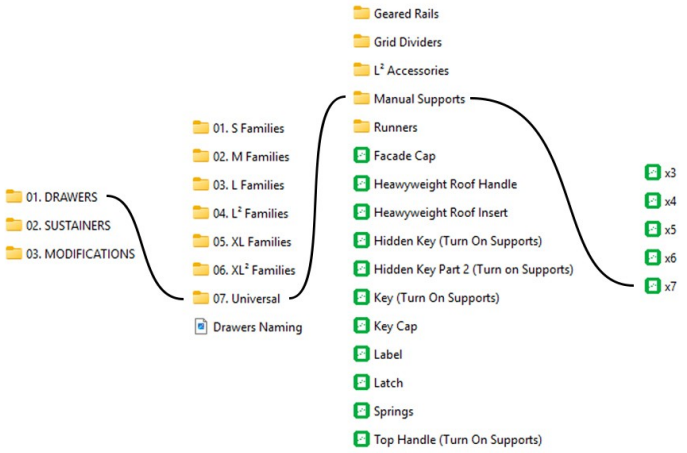
SCSAW (16d, 35D, 10mm thickness)

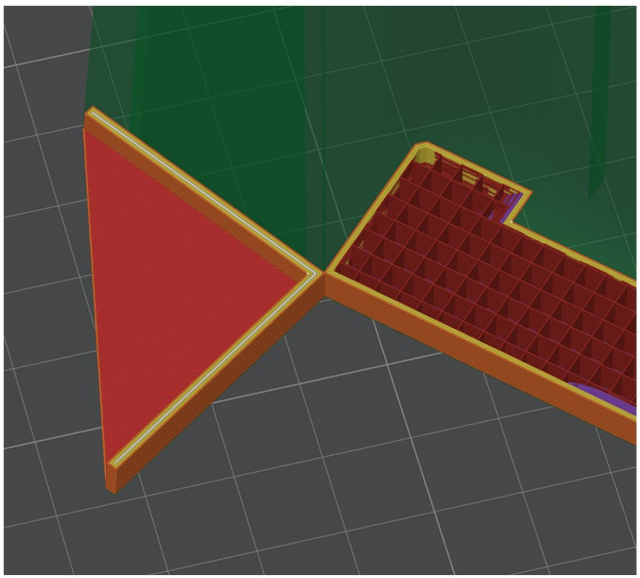
6205ZZ (25d, 52D, 15mm thickness)

Manual Supports









you can add manual supports for a more stable print experience, especially for tall prints